Password Policy Validation Engine

1. Problem Statement

Check a sample password against rules for minimum length, uppercase letter, lowercase letter, digit, and special character. Report every failed rule.

2. Objectives

Process strings using loops.

Check character categories.

Report multiple validation failures.

3. Assumptions

Minimum length is 8 characters.

A valid password must contain at least one uppercase letter, lowercase letter, digit, and special character.

Whitespace is not accepted as a special character.

4. Algorithm

1. Start.

2. Input a sample password.

3. Check its length.

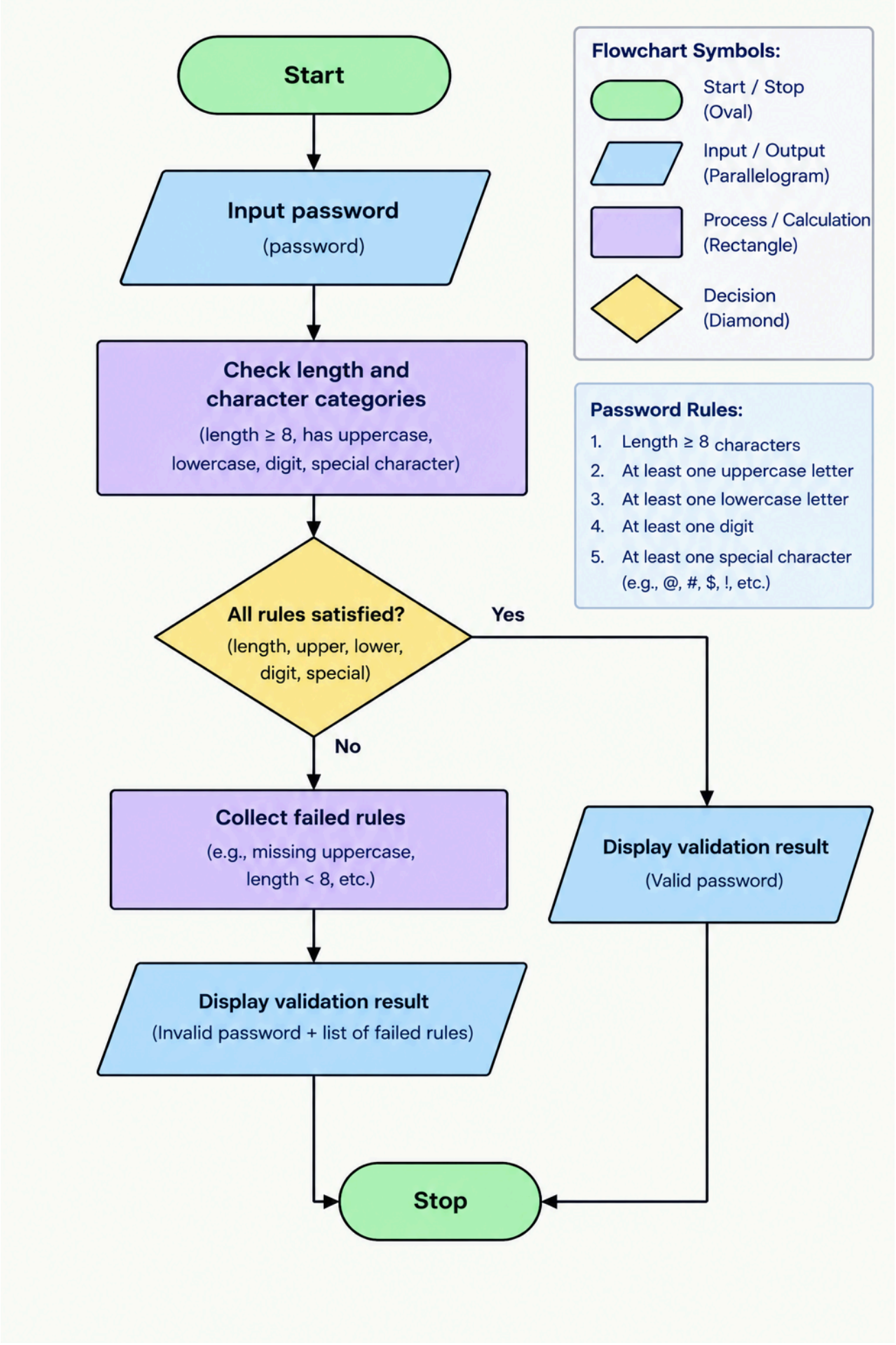
4. Loop through each character to identify uppercase, lowercase, digit, and special character.

5. Collect all failed rules.

6. Display whether the password passes and list failures.

7. Stop.

5. Flowchart (text form)



6. Pseudocode

START

INPUT password

Create flags for uppercase, lowercase, digit, special character

FOR each character: update the matching flag

Create empty list failures

IF length < 8: add length failure

IF any category flag is false: add corresponding failure

IF failures is empty: display valid; ELSE display invalid and failures

STOP

8. Test Case and Expected Output

Input: hello123

Password does not pass the sample policy. - Add at least one uppercase letter. - Add at least one special character.

9. Conclusion

The program checks a string against multiple password rules and explains each failed condition. It is a classroom example, not a complete real-world security checker.